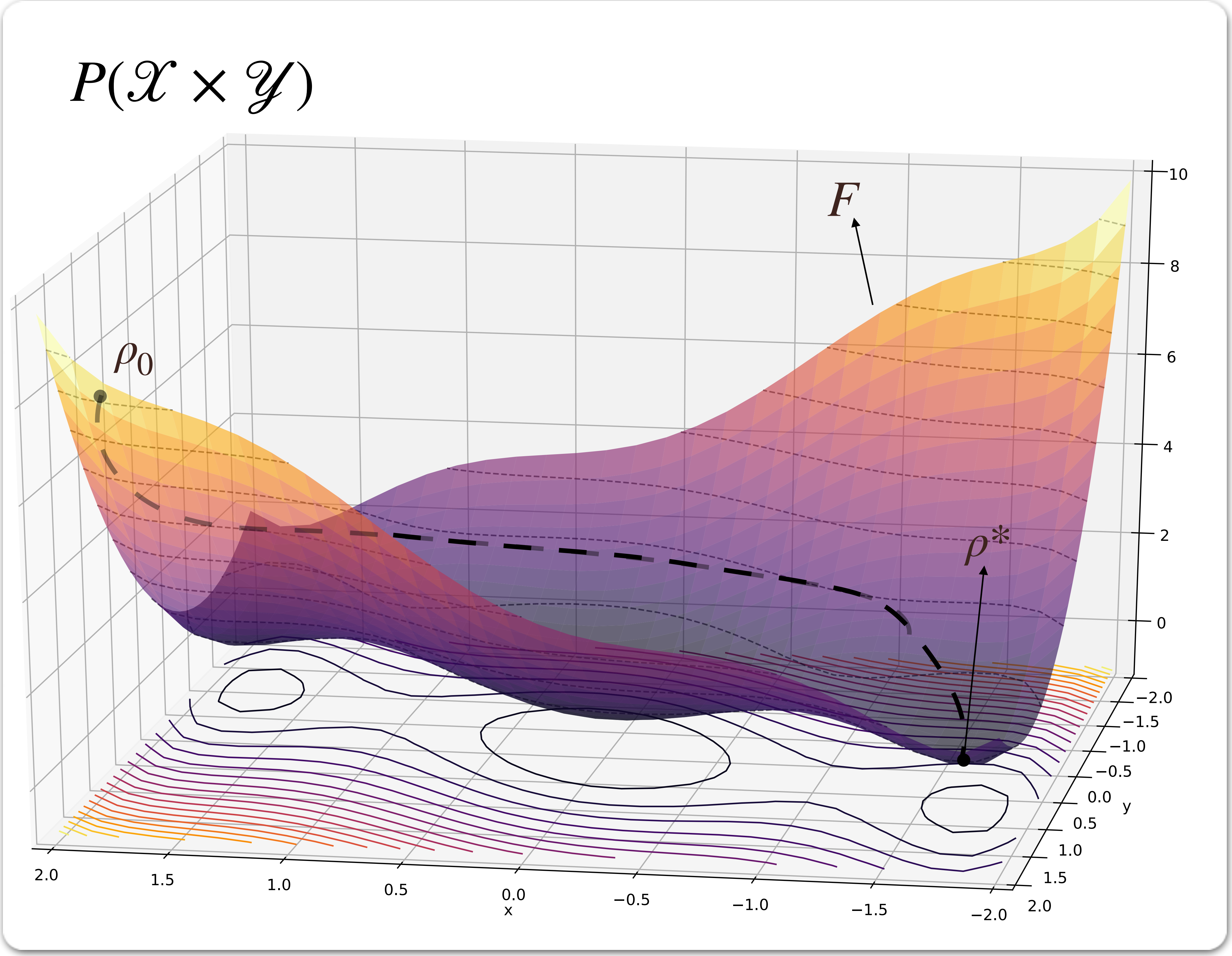
Parameter vector

$$x \in \mathcal{X}$$

Vector-valued function

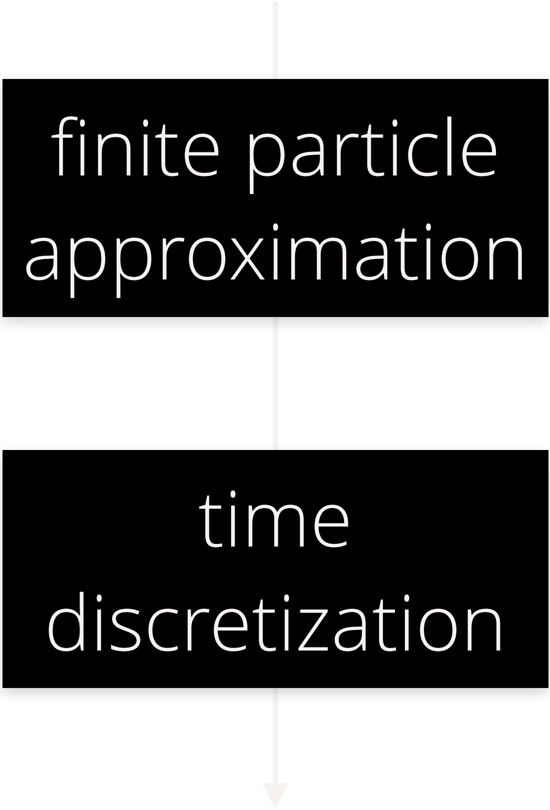
$$f : \mathcal{X} \to \mathbb{R}$$



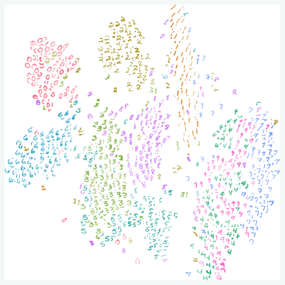
$P(\mathscr{X} \times \mathscr{Y})$
F
ρ_0
ρ^*
10
8
6
4
2
0
2.0
1.5
1.0
0.5
0.0
x
−0.5
−1.0
−1.5
−2.0
y

$$\partial_t \rho(t) = -\nabla \cdot \left(\rho(t) \nabla \frac{\delta F}{\delta \rho}(\rho(t)) \right), \qquad \rho(0) = \rho_0$$

$$z_{t+1}^{(i)} \leftarrow z_t^{(i)} - \lambda \nabla_{z^{(i)}} F(\hat{\rho}_t) \quad i = 1, \ldots, n$$


finite particle approximation
time discretization

Need to solve optimization problem over (**infinite dimensional**, **non-euclidean**!) probability space



Gradient flow in dataset space:

[Ambrosio; Santambrogio; Figalli; Villani; etc]

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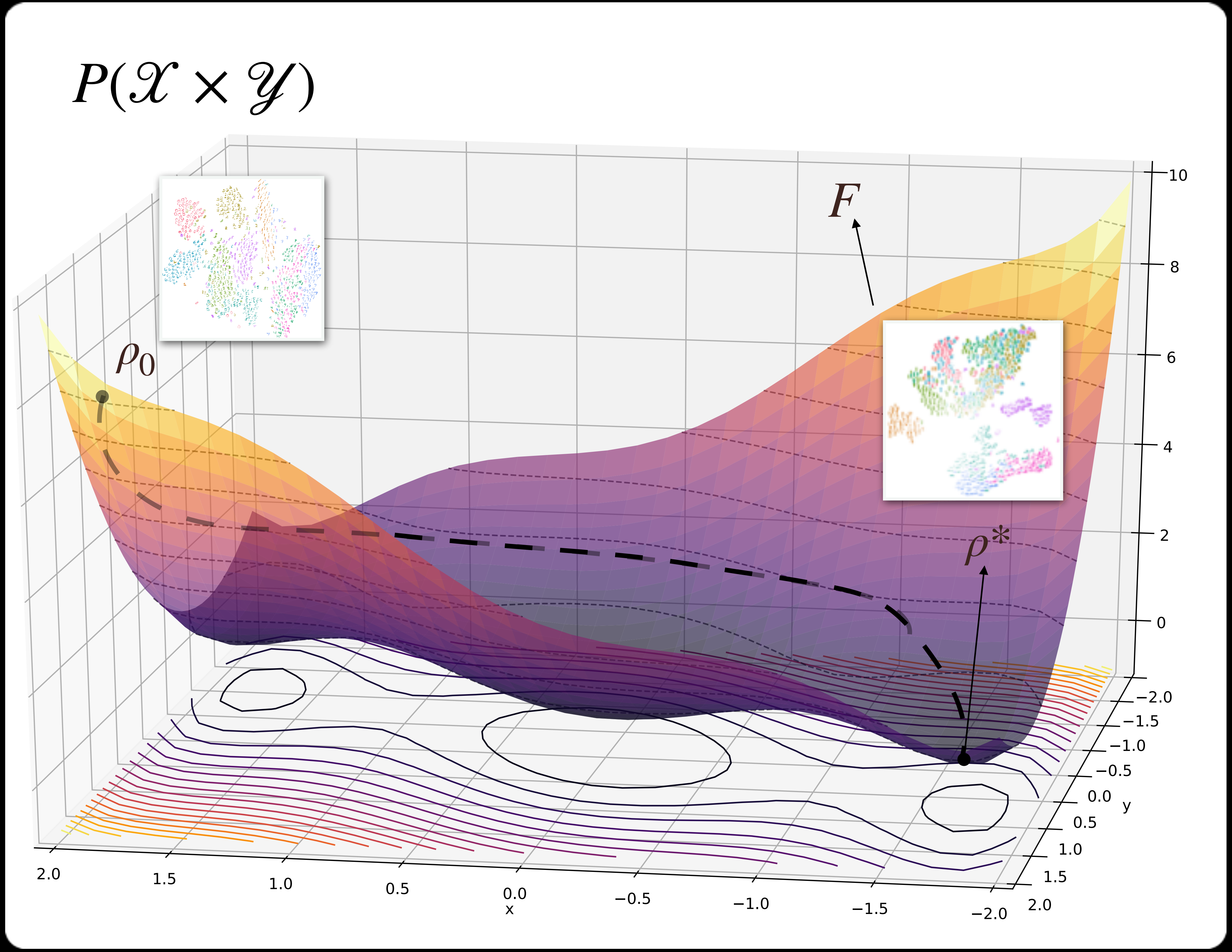
Dataset Optimization: Gradient Flows

Need to solve optimization problem over (**infinite dimensional**, **non-euclidean**!) probability space



Gradient flow in dataset space:

[Ambrosio; Santambrogio; Figalli; Villani; etc]

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finite particle approximation

time discretization

$$z_{t+1}^{(i)} \leftarrow z_t^{(i)} - \lambda \nabla_{z^{(i)}} F(\hat{\rho}_t) \quad i = 1, \ldots, n$$

each 'particle' is a data point (e.g., image)

Simple Dataset Dynamics

Objective: $F(\rho) = \mathrm{OTDD}(\rho, \beta)$